Madhab Chandra Das

Academic Resume

Huntsville, Texas | +1 (936) 438-3465 | mxd147@shsu.edu

Website | LinkedIn | GitHub | Google Scholar | ORCID | Scopus

Research Profile

Doctoral researcher and Graduate Research Assistant in Digital and Cyber Forensics at Sam Houston State University, with a current CGPA of 4.00. Combines forensic examination, Python scripting, and machine learning to study the reliability of digital evidence interpretation and AI-assisted reporting. Research spans Android, network and memory forensics, IoT security, and simulated quantum security. Supports undergraduate learning through lab instruction, assessment, and tutoring. Brings previous experience in banking due diligence, staff training, and team leadership.

Education

PhD in Digital and Cyber Forensics | Sam Houston State University

August 2025–Present | Expected July 2029 | Current CGPA: 4.00

Master's in Cybersecurity | Washington University of Science and Technology

October 2023–June 2025 | CGPA: 3.92/4.00

MBA | Jahangirnagar University | August 2020–July 2022

BSc in Electrical and Electronic Engineering | UITS | August 2008–January 2013

Academic Experience

Graduate Research Assistant and Teaching Assistant

Sam Houston State University, Department of Computer Science | August 2025–Present

- Conduct forensic research using Python, artifact examination, structured datasets, and evaluation of AI and machine learning outputs.
- Conduct lab classes, proctor examinations, grade exam responses, and provide tutoring outside scheduled class hours.

Courses supported

DFSC 3316 — Cryptography and Network Security | Spring and Fall 2026
COSC 1437 — Programming Fundamentals II | Fall 2025
DFSC 2316 — DF and IA Fundamentals II | Fall 2026

Research Projects

Android Forensic Reporting

Examined public Android 13 data and a physical LG device using Python and ALEAPP. Generated reports through three LLM APIs and evaluated 612 claims across six reports against verified findings.

Network Traffic Analysis

Prepared two encrypted traffic datasets and evaluated four ML models. Analyzed 41,238 external false positives (4.35% FPR) using SHAP and permutation importance.

Windows Memory Forensics

Built 1,899 process records with 18 features from 45 memory dumps using Volatility 2.6. Evaluated three ensembles with dump-grouped validation against rule-derived labels.

Quantum Security and Forensics

Generated two physics-inspired synthetic QKD datasets with 5 million records each. Implemented classifiers and examined reliability under changed simulation conditions.

Practical Forensic Training

Windows and Operating System Forensics: Forensic imaging, deleted file recovery, metadata and hex examination, password recovery, Registry and event log analysis, USB and browser artifacts, email headers, and HAR analysis.

Network Forensics: Wireshark and TShark analysis of TCP connections, handshakes, visible payloads, login-related information, and suspicious traffic.

Mobile and Memory Forensics: Android artifact examination with ALEAPP and memory analysis using Linux scripting.

E Discovery: Case exercises, record preservation, TAR, METS, and procedural rules. Tools: PST Analyzer, MBOX Viewer, GoldFynch.

Relevant Graduate Coursework

Operating System Forensics, Network Forensic Analysis, Live System & Memory Forensics, E-Discovery, Mobile Device Forensics, and Cryptography & Steganography.

Publications

Accepted

Madhab Chandra Das, Sajid Ahmed, Tasnia Tabassum, and Qingzhong Liu. “Evidence-Grounded LLM-Assisted Android Mobile Forensic Reporting.” IEEE TPS 2026. Accepted.

Madhab Chandra Das and Qingzhong Liu. “Explaining Cross-Dataset False Positives in Encrypted Traffic Classification for Forensic Interpretation.” IEEE UEMCON 2026. Accepted.

Madhab Chandra Das and Amar Rasheed. “Rule-Derived Process Classification in Windows Memory Forensics: A Reliability-Oriented Evaluation.” ISDFS 2027. Accepted.

Academic Service

Reviewer Service

CIDDF 2025: 3 papers. ICTA 2026: 5. QUANCOM 2026: 3. Cybersecurity, Springer Nature, 2026: 2 manuscripts. IEEE IRI 2026: 3 papers.

Program Committee Membership

QUANCOM 2026. IEEE ICTAI 2026. Chen Institute Symposium for AI Advancing Science & Society (AIAS^{+}), 2026.

Honors and Leadership

- COSET PhD Digital & Cyber Forensics Scholarship, SHSU, 2025–2026. Renewed for 2026–2027.
- WUST Presidential Honor: Summer 2024, Fall 2024, and Spring 2025.
- President, Bangladesh Student Association, SHSU | August 2025–Present. Organize events and support incoming students.

Professional Development

Cybersecurity and Risk Management Specialist — PIIT, 2024

Learning Cryptography and Network Security — LinkedIn Learning, 2026

Wireshark: Functionality — LinkedIn Learning, 2026

Certified Peer Reviewer Course — Elsevier Researcher Academy, 2025

Previous Professional Experience

The City Bank Ltd.: roles from SPMO and GL to Associate Manager, March 2017–July 2022. Mutual Trust Bank Ltd.: Senior Sales Officer, September 2016–February 2017. BRAC Bank Ltd.: Trainee Officer, June 2013–June 2016; Technology Intern, June–September 2011.